

Chapter 2) Kafka

- ① Zookeeper → managed by ensemble
Decided to keep odd number of nodes
As majority of consensus should win.
max 7 nodes as connected else performance degraded.
moving out of Zookeeper.
with Zookeeper Raft. (KIP-500)
Kafka Improvement Group

② Raft Mechanism.

Writes handled by leader.

Reads → can go to leader only for strong consistency.
can go to followers for state reads.

③ Broken id

Have their broken ids
unique.

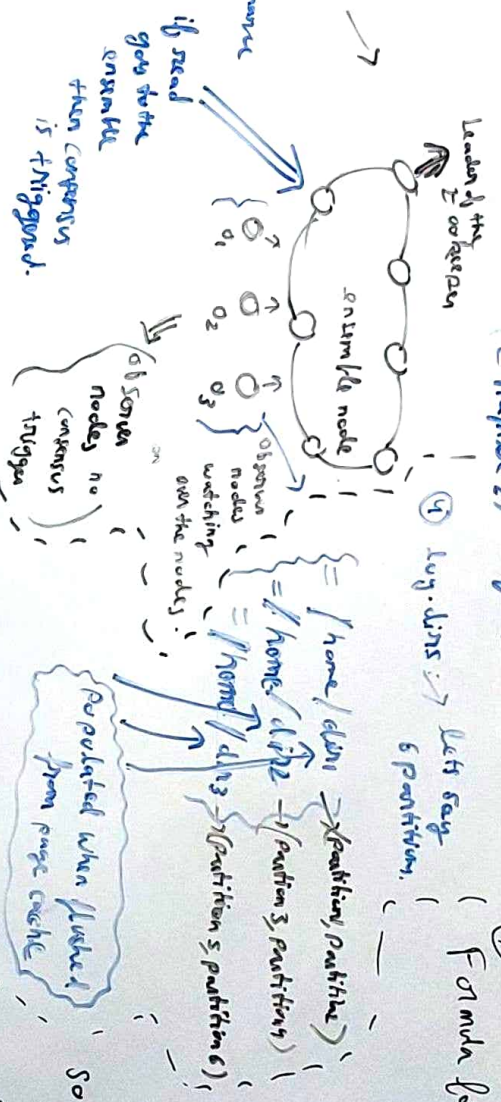
Connection info:

* hostname

* port

* path → connect path.

→ where Kafka
stores its metadata.



⑤ Data mostly stored in the page cache of the system.

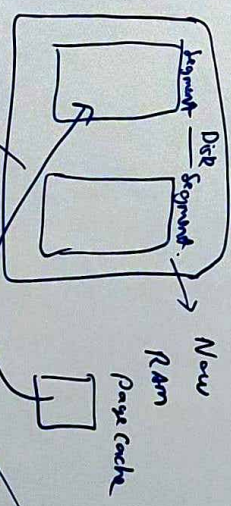
Partition Id.	Offset	Actual msg.

Background threads which sweep this buffer every 100ms.

When consumer comes to read it checks if present in the page cache. If yes then read from here, or go to disk.

⑧

In OS at the disk level, how these partitions are flushed is important.



log segment bytes → often how many bytes should segment be closed.

After how many seconds should segment be closed.

⑥. New config
auto. Rebal. on kafka → this should auto 1.
if a kafka node leaves the leader for multiple topics' partition, then we are creating a single point of failure in us.

So if expected message rate is 120GB/day

$$\# = \left(\frac{120}{6} \right) = 20 \text{ partitions.}$$

this is approximated as 6GB/day.

Chapter 2) Kafka.

① We know Kafka is written in Java, hence it is allocated heap memory, Java CPU.

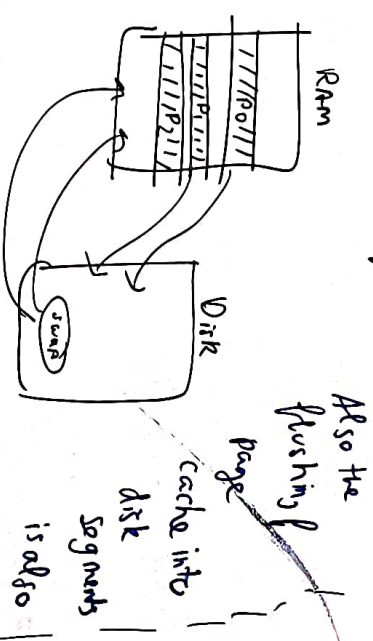
So it is recommended that on the broker only host the Kafka application itself and nothing else. To provide it higher heap size, higher page cache size.

② Kafka in cloud.

Kafka has AWS or Azure managed.

HDDs → cheap but don't have tight SLA
SSDs → costly but 99.999% SLA.

③ We know how OS handles the paging stuff.



④ Handle dirty pages

Dirty pages are pages which are there on the page cache and get to be flushed.

* (VM-dirty-background-ratio = 5%)

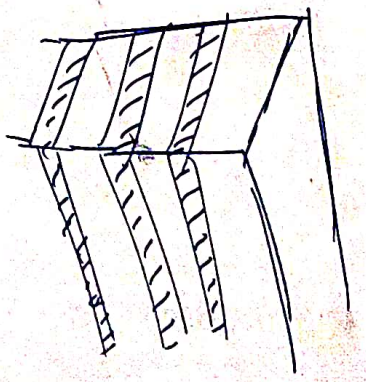
this tells how much % of Ram if it gets filled should a background thread start writing to disk.

$$* \text{VM-dirty-ratio} = (60 - 50)\%$$

this tells how much % of Ram if it gets filled then start a flushing thread in the main thread and flush the pages.

⑤ Lastly Rack management

if for a partition, all the ISRs are in the same rack and if the RACK goes down, then the entire partition is down.



⑥ In Zookeeper, it's sometimes difficult to keep 1 node in the ensemble for leader consensus.

And for each broker on topic

/chroot1 → Different folders can be used as metadata storage for each topic
/chroot2 →